# 2026 VCE Mathematical Methods 1 (NHT) external assessment report

This report provides sample answers, or an indication of what answers may have included. Unless otherwise stated, these are not intended to be exemplary or complete responses.

## Question 1a.

This question involved using the product rule for differentiation.

$$\frac{dy}{dx} = 3x^2 \log_e(5x) + x^3 \times \frac{1}{5x} \times 5$$
$$= 3x^2 \log_e(5x) + x^2 = x^2(3\log_e(5x) + 1)$$

## Question 1b.

This question could be solved by applying the integration rule from the formula sheet, or by expanding the expression first and then finding the antiderivative. The resulting antiderivative expression then needed to be evaluated using the boundary condition $f(1) = 1$.

### Method 1

$$f(x) = \int (x-2)^3 \, dx = \frac{1}{4}(x-2)^4 + c$$

$$f(1) = 1 \Rightarrow \frac{1}{4} + c = 1, \; c = \frac{3}{4}$$

$$\therefore \quad f(x) = \frac{1}{4}(x-2)^4 + \frac{3}{4}$$

### Method 2 – Expansion and integration

$f'(x) = x^3 - 6x^2 + 12x - 8$

$f(x) = \frac{x^4}{4} - 2x^3 + 6x^2 - 8x + d$

$f(1) = 1 \Rightarrow \frac{1}{4} - 2 + 6 - 8 + d = 1, \quad d - \frac{15}{4} = 1, \quad d = \frac{19}{4}$

$\therefore \quad f(x) = \frac{x^4}{4} - 2x^3 + 6x^2 - 8x + \frac{19}{4}$

## Question 2

This question involved using the information given to determine the value of the unknowns $a$, $m$ and $c$. The axial intercepts could be used to set up a pair of simultaneous equations that could be solved to find the value of $m$ and $c$.

A vertical asymptote at $x = -6 \Rightarrow a = 6$

$f(x) = \frac{m}{x+6} + c$

Using the axial intercepts $(-8, 0)$ & $(0, 24)$

$f(-8) = 0 \Rightarrow -\frac{m}{2} + c = 0 \Rightarrow c = \frac{m}{2} \quad ...(1)$

$f(0) = 24 \Rightarrow \frac{m}{6} + c = 24 \quad ...(2)$

Sub (1) into (2):

$\frac{m}{6} + \frac{m}{2} = 24 \Rightarrow \frac{2m}{3} = 24 \Rightarrow m = 36, c = 18$

## Question 3a.

The probability value could be expressed in any correct form.

$\frac{85}{100} = \frac{17}{20} = 0.85 (= 85\%)$

## Question 3b.

This question involved calculating the mean or expected value of a discrete probability distribution using the formula from the formula sheet.

Let $X$ be the random variable for the price of a ticket.

$$\begin{aligned}\mathrm{E}(X) &= 5\times 0.60 + 8\times 0.25 + 12\times 0.15 \\ &= 3.00 + 2.00 + 1.80 \\ &= \$\,6.80 \quad (6 \text{ dollars } 80 \text{ cents},\ 6.8,\ \frac{34}{5})\end{aligned}$$

## Question 3c.

This question required the identification and construction of the binomial probability distribution for the number of visitors purchasing a ticket for less than 10 minutes, $Y \sim \mathrm{Bi}(n = 5, p = 0.6)$. The probability that exactly 4 out of 5 visitors buy a ticket needed to be found, and this could be calculated using the appropriate formula provided on the formula sheet.

Let $Y$ be the number of visitors from the random sample that spent less than 10 minutes at the location.

$$\begin{aligned}\Pr(Y=4) &= \binom{5}{4}(0.6)^4(0.4)^1 \\ &= 5\times\left(\frac{3}{5}\right)^4\times\frac{2}{5} \\ &= \begin{cases}\dfrac{2\times 6^4}{10^4} = \dfrac{2\times 3^4}{5^4} \\ \dfrac{810}{3125} = \dfrac{162}{625} (\text{or } 0.2592)\end{cases}\end{aligned}$$

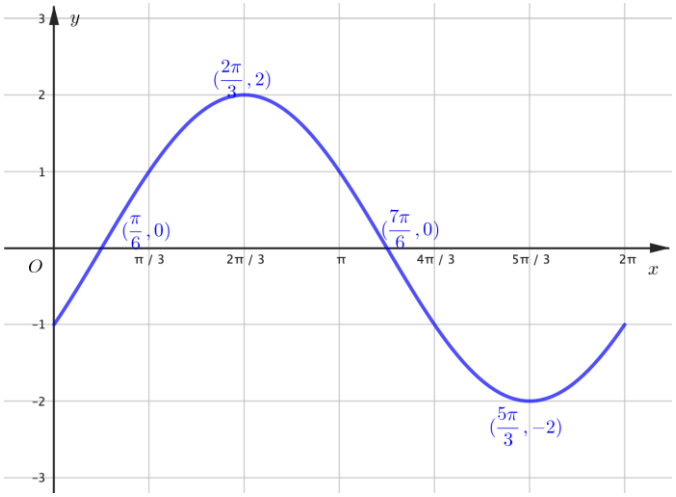
## Question 4a.

This question involved the calculation of the value of the function when $x = 0$.

$f(0) = -1$

## Question 4b.

The graph of the function needed to be sketched over the domain $[0,2\pi]$. The shape needed to represent a sine graph depicting a period of $2\pi$ and an amplitude of $2$. Endpoints, although not required to be labelled, needed to be correctly positioned. The two $x$-intercepts needed to be labelled and correctly positioned. The turning points needed to be correctly positioned and labelled.




## Question 5a.

The question involved finding the equations of all vertical asymptotes over the maximal domain $R$. The equation needed to be written as $x =$ with the introduced parameter $k$ (or $n$) defined as $k \in Z$. The equations could be expressed in any correct form.

$$x = \frac{\pi}{4} + \frac{k\pi}{2}, k \in Z$$

$$x = \frac{\pi}{4}(1 + 2k), k \in Z$$

$$x = \frac{\pi}{4}(\pm 1 \pm 2k), \ k \in Z$$

## Question 5b.

This question involved using the chain rule to differentiate the tangent function, the formula for which can be found on the formula sheet.

$$f'(x) = 2\sec^2(2x) = \frac{2}{\cos^2(2x)}$$

## Question 5c.

This question involved the calculation of the area bounded by the graph of $y = f'(x)$, the $x$-axis and the lines $x = \frac{\pi}{12}$ and $x = \frac{\pi}{6}$. Antidifferentiating $f'(x)$ gave $f(x)$ and then the integral needed to be evaluated at the correct terminals of $\frac{\pi}{12}$ and $\frac{\pi}{6}$. The answer needed to be expressed as a single term.

A graph of $y = f'(x)$ can be considered over $x \in \left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$. Note that $f'(x) \geq 2 > 0$, so the region has a positive area above the $x$-axis.

$$
\begin{aligned}
A &= \int_{\frac{\pi}{12}}^{\frac{\pi}{6}} 2\sec^2(2x)\,dx \\
&= \left[\tan(2x)\right]_{\frac{\pi}{12}}^{\frac{\pi}{6}} \\
&= \tan\left(\frac{\pi}{3}\right) - \tan\left(\frac{\pi}{6}\right) \\
&= \sqrt{3} - \frac{\sqrt{3}}{3} \quad \left(\text{or } \sqrt{3} - \frac{1}{\sqrt{3}}\right) \\
&= \frac{2\sqrt{3}}{3} \quad \left(\text{or } \frac{2}{\sqrt{3}}\right)
\end{aligned}
$$

or

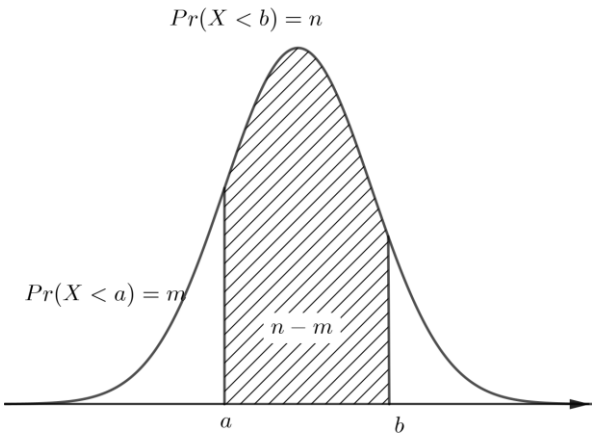
$$
\begin{aligned}
A &= \int_{\frac{\pi}{12}}^{\frac{\pi}{6}} f'(x)\,dx \\
&= \left[f(x)\right]_{\frac{\pi}{12}}^{\frac{\pi}{6}} \\
&= f\left(\frac{\pi}{6}\right) - f\left(\frac{\pi}{12}\right) \\
&= \sqrt{3} - \frac{\sqrt{3}}{3} \\
&= \frac{2\sqrt{3}}{3} \quad \left(\text{or } \frac{2}{\sqrt{3}}\right)
\end{aligned}
$$

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## Question 6

This question involved the calculation of a conditional probability. Initially there needed to be evidence of the calculation of either $\Pr(a < X < b)$ on the numerator or $\Pr(X < b)$ on the denominator. A graph could be drawn to indicate the required intersection. The answer could be given in either form.

Working to find the numerator:




$$\Pr(X > a \mid X < b) = \frac{\Pr(a < X < b)}{\Pr(X < b)}$$
$$= \frac{\Pr(X < b) - \Pr(X < a)}{\Pr(X < b)}$$
$$= \frac{n-m}{n} \quad \left(\text{or } 1 - \frac{m}{n}\right)$$

## Question 7a.

This question was a 'show that' question. As such, each line of working needed to demonstrate a clear, logical and explicit progression, leading to the answer. Specifically, it was important to show how the manipulation of $k$ led directly to $\sqrt{k^2}$.

$$A(k) = k \times f(k) = k \times \sqrt{12 - k}$$
$$= \sqrt{k^2}\sqrt{12 - k}$$
$$= \sqrt{k^2(12 - k)}$$
$$= \sqrt{12k^2 - k^3}$$

# Question 7b.

There were different ways to approach this question. The derivative could be found using the chain rule on the function $A(k)=\sqrt{12k^2-k^3}$ given in part 7a., the derivative could be found using the product rule on the function $A(k)=k\sqrt{12-k}$, or the function $A(k)^2$ could be constructed and differentiated. In any case, the derivative needed to be set equal to zero and solved to find the value of $k$. The domain was restricted to $(0,12)$ so the only solution was $k=8$.

## Method 1

$$A'(k)=\frac{24k-3k^2}{2\sqrt{12k^2-k^3}}=\frac{3\left(8k-k^2\right)}{2\sqrt{12k^2-k^3}}=0 \text{ for maximum}$$

OR

$$A(k)=\left(12k^2-k^3\right)^{\frac{1}{2}},\ A'(k)=\frac{1}{2}\left(24k-3k^2\right)\left(12k^2-k^3\right)^{-\frac{1}{2}}$$

$A'(k)=0$ for maximum

$8k-k^2=0\Rightarrow k\left(8-k\right)=0\Rightarrow k=0,\ k=8$

But $0<k<8$

$\therefore\quad k=8$

## Method 2

$$A(k)=k\sqrt{12-k}$$

$$A'(k)=\sqrt{12-k}-\frac{k}{2\sqrt{12-k}}=\frac{3(8-k)}{2\sqrt{12-k}}=0$$

$A'(k)=0$ when $k=8$

## Method 3

$A\left(k\right)$ is a maximum when $A\left(k\right)^2$ is a maximum

$$A\left(k\right)^2=12k^2-k^3$$

$$\frac{d}{dk}\left(A\left(k\right)^2\right)=24k-3k^2$$

$$\frac{d}{dk}\left(A\left(k\right)^2\right)=0, \text{when } 24k-3k^2=0, \text{ so } k=8$$

## Question 8a.

This question required rearrangement of the terms and use of logarithm laws to arrive at a quadratic equation in terms of $x$. The quadratic equation could be readily solved and symmetry could be used to find the $y$-value of $y = 1$ for both points.

$$\log_e\left(1+x^2\right) = 2 - \log_e\left(1+x^2\right)$$

$$\Rightarrow 2\log_e\left(1+x^2\right) = 2,\ \log_e\left(1+x^2\right) = 1$$
$$\Rightarrow 1+x^2 = e,\ x^2 = e-1$$
$$\Rightarrow x = \pm\sqrt{e-1}$$

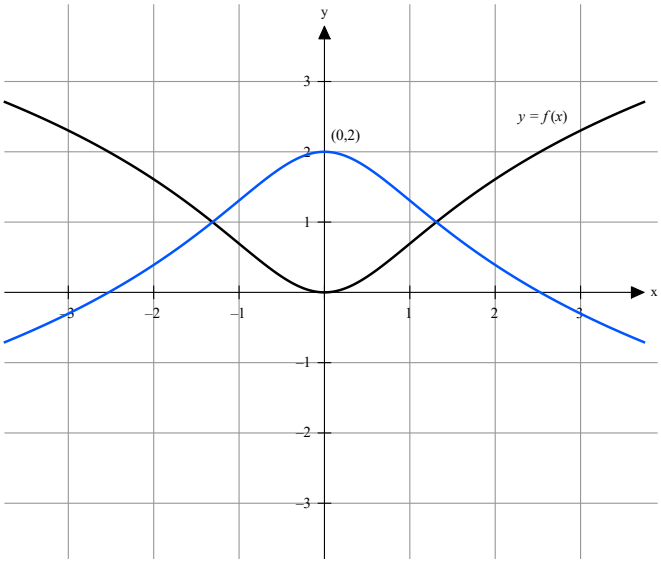
Then, by symmetry, $f\left(\sqrt{e-1}\right) = f\left(-\sqrt{e-1}\right)$

Note that $x^2 = e-1$ and $y = \log_e\left(1+e-1\right) = 1$

$$\therefore\ \left(\sqrt{e-1}, 1\right) \& \left(-\sqrt{e-1}, 1\right)$$

## Question 8b.

This question required students to sketch the graph of the transformed function $g(x) = 2 - f(x)$, whereby the graph of $y = f(x)$ is reflected in the $x$-axis and then translated up 2 units. The resulting graph needed to look like a reflection of the original graph in the line $y = 1$, with $x$-intercepts at approximately $x = \pm 2.5$, a $y$-intercept labelled at $(0, 2)$ and intersection points with the graph of $y = f(x)$ at $y = 1$.



## Question 8c.

This question involved transforming the point $\left(\sqrt{2}, \log_e(3)\right)$ according to a sequence of transformations.

Students chose to either transform the point directly or construct a set of mapping rules for $x'$ and $y'$ to find the transformed point.

$\left(\sqrt{2}+3, \frac{1}{2}\log_e(3)\right)$ or $\left(\sqrt{2}+3, \log_e\left(\sqrt{3}\right)\right)$

## Question 9a.

This question was a 'show that' question. As such, each line of working needed to demonstrate a clear, logical and explicit progression, leading to the answer. Students could choose to separate out the terms and then differentiate and recombine, or they could use the product rule or quotient rule directly to produce the derivative. The derivative needed to be given in the form as stated in the question.

### Method 1

Separate terms, differentiate and combine.

$$f(x) = \frac{2}{e^{2x}} - \frac{e^x}{e^{2x}} = 2e^{-2x} - e^{-x}$$

$$f'(x) = -4e^{-2x} + e^{-x}$$

$$= \frac{1}{e^x} - \frac{4}{e^{2x}} = \frac{1}{e^x} \times \frac{e^x}{e^x} - \frac{4}{e^{2x}}$$

$$= \frac{e^x - 4}{e^{2x}}$$

### Method 2

Use the quotient rule directly.

$$f'(x) = \frac{-e^x\left(e^{2x}\right) - \left(2 - e^x\right) \cdot 2e^{2x}}{\left(e^{2x}\right)^2}$$

$$= \frac{-e^{3x} - 4e^{2x} + 2e^{3x}}{e^{4x}}$$

$$= \frac{e^{3x} - 4e^{2x}}{e^{4x}}$$

$$= \frac{e^x - 4}{e^{2x}}$$

### Method 3

Use the product rule directly.

$$
\begin{aligned}
f(x) &= e^{-2x}\left(2-e^{x}\right) \\
f'(x) &= -2e^{-2x}\left(2-e^{x}\right)+e^{-2x}\left(-e^{x}\right) \\
&= e^{-2x}\left(-4+2e^{x}-e^{x}\right) \\
&= \frac{1}{e^{2x}}\left(e^{x}-4\right) \\
&= \frac{\left(e^{x}-4\right)}{e^{2x}}
\end{aligned}
$$

## Question 9bi.

This question involved setting the derivative equal to zero and solving to find the coordinates of the stationary point.

$$f'(x)=0 \Rightarrow e^{x}-4=0 \Rightarrow x=\log_e(4)$$

(Alternatively, $2e^{-2x}-e^{-x}=0$)

$$f\left(\log_e(4)\right)=\frac{2-4}{4^2}=-\frac{1}{8}$$

The stationary point is $\left(\log_e(4), -\frac{1}{8}\right)$

or using full substitution

$$
\begin{aligned}
f\left(\log_e(4)\right) &= \frac{2-e^{\log_e(4)}}{\left(e^{\log_e(4)}\right)^2} \\
&= \frac{2-4}{e^{2\log_e(4)}} = -\frac{2}{e^{\log_e(16)}} = -\frac{2}{16} = -\frac{1}{8}
\end{aligned}
$$

## Question 9bii.

There were different ways that this question could be approached. Students could test the derivative either side of the point, they could use the second derivative test or they could test the $y$-values either side of the point. It was important to see that actual calculations were made and that conclusions were drawn from actual calculations.

## Method 1 – First derivative test (sub $x$-values)

For example, choose $x_1 = \log_e(2) < \log_e(4)$ and $x_2 = \log_e(6) > \log_e(4)$

$$f'\left(\log_e(2)\right) = \frac{2-4}{2^2} = -\frac{1}{2} < 0$$
$$f'\left(\log_e(6)\right) = \frac{6-4}{6^2} = \frac{1}{18} > 0$$

Therefore, the point $\left(\log_e(4), -\frac{1}{8}\right)$ is a local minimum (or minimum / global minimum).

Or other points:

| $x$ | $\log_e(3)$ | $\log_e(4)$ | $\log_e(5)$ |
|---|---|---|---|
| $f'(x)$ | $\frac{-1}{9} < 0$ | $0$ | $\frac{1}{25} > 0$ |

The stationary point is a minimum turning point.

## Method 2 – Second derivative test

$$f'(x) = e^{-x} - 4e^{-2x}$$
$$f''(x) = -e^{-x} + 8e^{-2x}$$
$$f''\left(\log_e(4)\right) = \frac{1}{4} > 0$$

At $x = \log_e(4)$, the graph is concave up and there is a local minimum (or minimum / global minimum).

## Method 3 – Testing $y$-values either side

For example, choose $x = 0$ and $x = \log_e(5)$

At $x = 0$, $f(0) = 1 > \frac{-1}{8}$

At $x = \log_e(5)$, $f(\log_e 5) = \frac{-3}{25} > \frac{-1}{8}$

Since $f(\log_e 5) > f(\log_e 4)$ and $f(0) > f(\log_e 4)$ stationary point is a minimum.

Hence, the point $\left(\log_e(4), -\frac{1}{8}\right)$ is a local minimum (or minimum / global minimum).

## Question 9c.

This question required students to find the values of $x$ such that $f(x) \geq 0$.

$f(x) \geq 0 \Rightarrow \frac{2-e^x}{e^{2x}} \geq 0$, but $e^{2x} > 0$ for $x \in R$

Thus,

$e^x \leq 2$
$\Rightarrow x \leq \log_e(2)$ or $x \in (-\infty, \log_e(2)]$

## Question 9d.

This question involved finding the integral of $f(x) = \frac{2-e^x}{e^{2x}}$ and evaluating at the terminals $x = q$ and $x = p$ in order to obtain an expression for $p$ in terms of $q$. The resulting equation could then be factorised and solved by using either a difference of two squares approach, completing the square, or by forming a quadratic in terms of $e^{-p}$ and using the quadratic formula. It was important to discard the case when $p = q$.

### Method 1

$$\int_p^q \frac{2-e^x}{e^{2x}}dx = \int_p^q 2e^{-2x} - e^{-x}dx$$
$$= \left[-e^{-2x} + e^{-x}\right]_p^q$$
$$= \left(-e^{-2q} + e^{-q}\right) - \left(-e^{-2p} + e^{-p}\right) = 0$$

$e^{-p} - e^{-q} = e^{-2p} - e^{-2q}$

$e^{-p} - e^{-q} = \left(e^{-p} - e^{-q}\right)\left(e^{-p} + e^{-q}\right)$

as $p \neq q$, $e^{-p} - e^{-q} \neq 0$

$\therefore\ 1 = e^{-p} + e^{-q}$

$e^{-p} = 1 - e^{-q}$

$-p = \log_e\left(1 - e^{-q}\right)$

$\therefore p = -\log_e\left(1 - e^{-q}\right)$ or $p = \log_e\left(\frac{e^q}{e^q - 1}\right)$ for $q > 0$

$$\therefore \quad p = \begin{cases} \log_e\left(\dfrac{e^q}{e^q-1}\right) \\ \log_e\left(\dfrac{1}{1-e^{-q}}\right) \\ q-\log_e\left(e^q-1\right) \\ \log_e\left(e^q\right)-\log_e\left(e^q-1\right) \\ q+\log_e\left(\dfrac{1}{e^q-1}\right) \\ -\log_e\left(1-e^{-q}\right) \end{cases}$$

## Method 2

$$\begin{aligned}\int_p^q \frac{2-e^x}{e^{2x}}dx &= \int_p^q 2e^{-2x}-e^{-x}dx \\ &= \left[-e^{-2x}+e^{-x}\right]_p^q \\ &= \left(-e^{-2q}+e^{-q}\right)-\left(-e^{-2p}+e^{-p}\right)=0\end{aligned}$$

$$e^{-2q}-e^{-q} = e^{-2p}-e^{-p}$$

$$e^{-2q}-e^{-q}+\frac{1}{4} = e^{-2p}-e^{-p}+\frac{1}{4}$$

$$\left(e^{-q}-\frac{1}{2}\right)^2 = \left(e^{-p}-\frac{1}{2}\right)^2$$

$$\left(e^{-q}-\frac{1}{2}\right) = \begin{cases} \left(e^{-p}-\dfrac{1}{2}\right), \text{ reject this case as } p \neq q \\ -\left(e^{-p}-\dfrac{1}{2}\right) \end{cases}$$

$$\therefore e^{-q}-\frac{1}{2} = -e^{-p}+\frac{1}{2}$$

$$e^{-p} = -e^{-q}+1$$

$$-p = \log_e(1-e^{-q})$$

$$p = -\log_e(1-e^{-q})$$

## Method 3

$$\begin{aligned}\int_p^q \frac{2-e^x}{e^{2x}}dx &= \int_p^q 2e^{-2x}-e^{-x}dx\\ &= \left[-e^{-2x}+e^{-x}\right]_p^q\\ &= \left(-e^{-2q}+e^{-q}\right)-\left(-e^{2p}+e^{-p}\right)\\ &= \frac{1}{e^{2p}}-\frac{1}{e^{2q}}+\frac{1}{e^q}-\frac{1}{e^p}\end{aligned}$$

$$\begin{aligned}\int_p^q f(x)dx=0 &\Rightarrow \frac{1}{e^{2p}}-\frac{1}{e^{2q}}+\frac{1}{e^q}-\frac{1}{e^p}=0\\ &\Rightarrow \frac{1}{e^{2p}}-\frac{1}{e^{2q}}=\frac{1}{e^p}-\frac{1}{e^q}\\ &\Rightarrow \frac{e^{2q}-e^{2p}}{e^{2p}\cdot e^{2q}}=\frac{e^q-e^p}{e^p\cdot e^q}\\ &\Rightarrow \frac{e^{2q}-e^{2p}}{e^{2p}\cdot e^{2q}}=\frac{e^q-e^p}{e^p\cdot e^q}\cdot\frac{e^pe^q}{e^pe^q}\\ &\Rightarrow e^{2q}-e^{2p}=e^pe^q\left(e^q-e^p\right)\quad\left(\text{cancel as } e^p\neq e^q\right)\\ &\Rightarrow \left(e^q+e^p\right)\left(e^q-e^p\right)=e^pe^q\left(e^q-e^p\right)\\ &\Rightarrow e^q+e^p=e^pe^q,\quad e^pe^q-e^p=e^q\\ &\Rightarrow e^p\left(e^q-1\right)=e^q\\ &\Rightarrow e^p=\frac{e^q}{e^q-1}\end{aligned}$$

$$\therefore\quad p=\begin{cases}\log_e\left(\dfrac{e^q}{e^q-1}\right)\\ \log_e\left(\dfrac{1}{1-e^{-q}}\right)\\ q-\log_e\left(e^q-1\right)\\ \log_e\left(e^q\right)-\log_e\left(e^q-1\right)\\ q+\log_e\left(\dfrac{1}{e^q-1}\right)\\ -\log_e\left(1-e^{-q}\right)\end{cases}$$